



The Caillou Project

Internship project May-August 2026



Context



The upcoming AMADEE-27 mission, organized by the Austrian Space Forum (OeWF), represents a major milestone in high-fidelity Mars analog simulations. Based in Portugal, the mission will test key instruments and exploration workflows, critical to the geological search for life on Mars. In planetary science, Raman spectroscopy is widely recognized as an innovative and highly effective non-destructive method for identifying minerals and potential bio-markers in situ. However, deploying real space-grade instruments in the field is too complex and expensive.

To bridge the gap between computational models and real-world field acquisition, a team of ÖWF's intern focuses on developing "Caillou," a functional prototype of a Synthetic Raman Spectrometer. Rather than performing authentic chemical analyses, this transportable guest-payload is specifically engineered to replicate the complete operational experience during an Extravehicular Activity (EVA). By mirroring the entire operational workflow, the Caillou project provides an adaptable, cost-effective platform to evaluate mission procedures, refine scientific decision-making, and prepare future explorers for the rigorous constraints of Martian exploration.



Content



 **System Requirements & Operational Scenario**

 **Hardware & Mechanical Design**

 **Software & Data Workflow *(In Progress)***

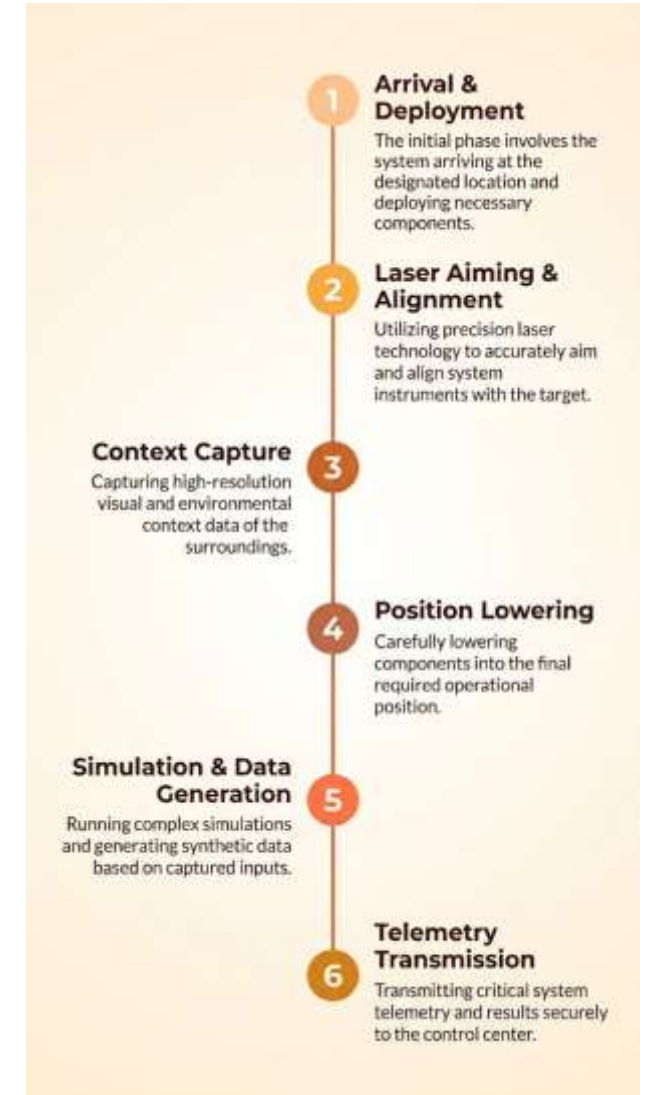


System Requirements & Operational Scenario



Functional & Data Workflow

- **Realistic Workflow:** Replicate the complete setup and data path of a space-grade spectrometer without actual chemical analysis
- **Targeting & Context:** Integrate a visible laser for aiming and an onboard camera to capture high-resolution sample images
- **Synthetic Output:** Generate physically plausible spectra from a local library and transmit them seamlessly to the Science Support Team (SST)



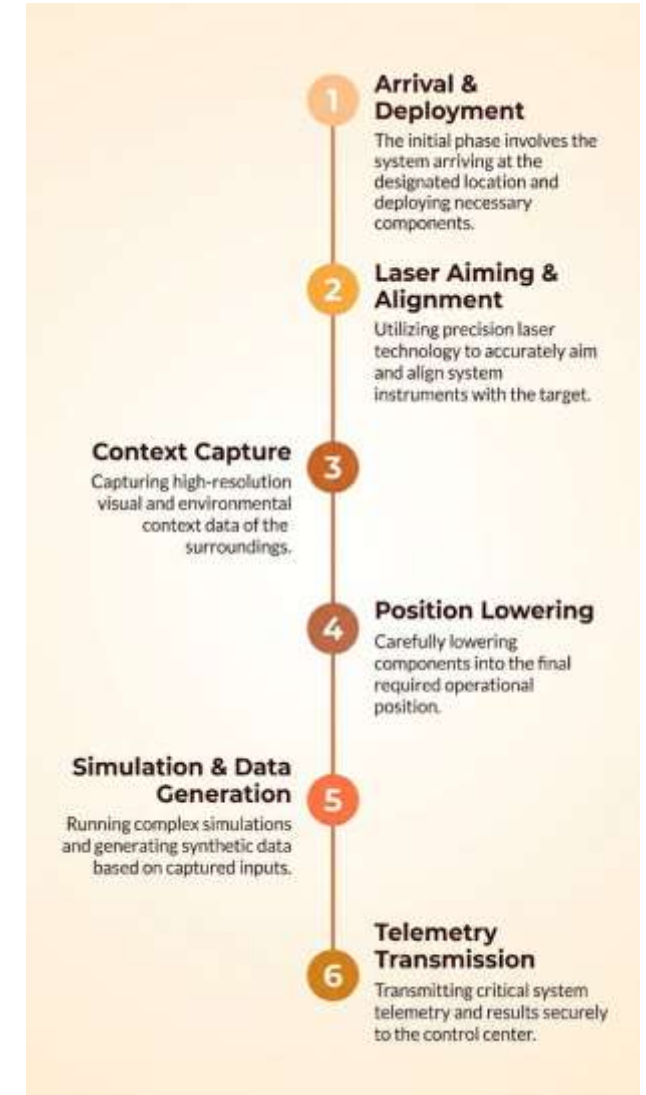


System Requirements & Operational Scenario



Mechanical & Ergonomics

- **Modular Interface:** Adaptable mounting system to switch quickly between a rover mount and manual handling by an astronaut
- **EVA-Friendly:** Compact, lightweight, and robust enough to be operated effortlessly with heavy spacesuit gloves
- **Field Durability:** Custom enclosure designed to withstand dust, transport, and analog field conditions



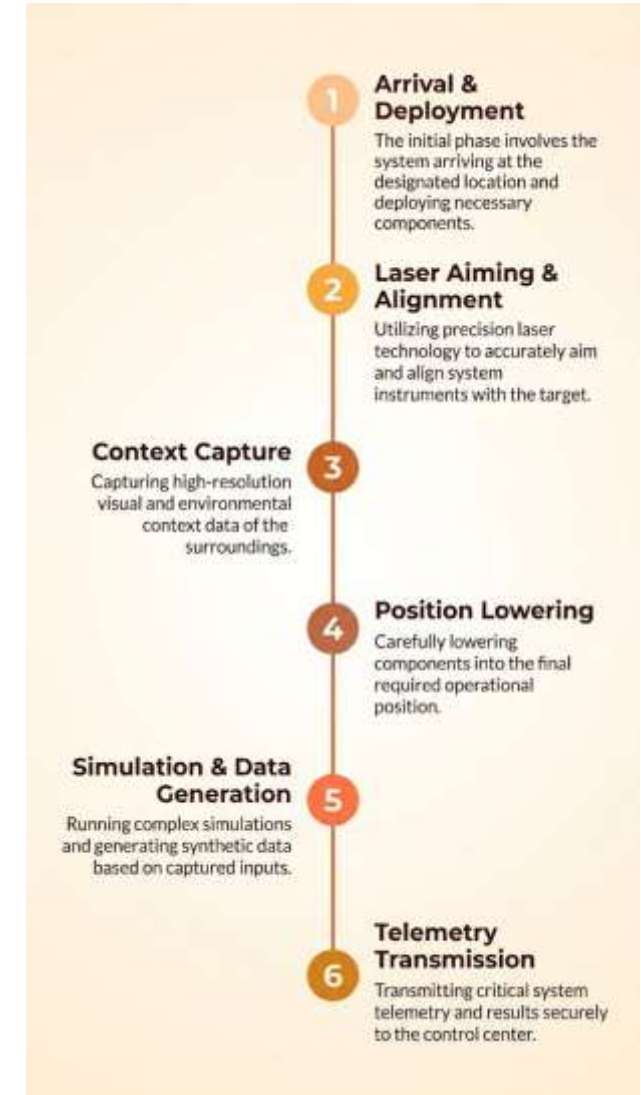


System Requirements & Operational Scenario



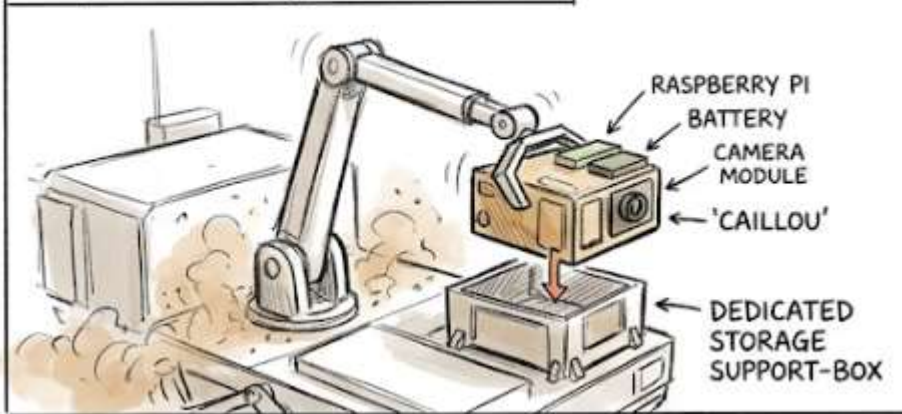
Electronics & Power

- **SBC Core:** Powered by a versatile Raspberry Pi to control the laser, camera, and data management
- **Power Autonomy:** Operated via a lightweight, commercial power bank ensuring 2+ hours of continuous EVA operations

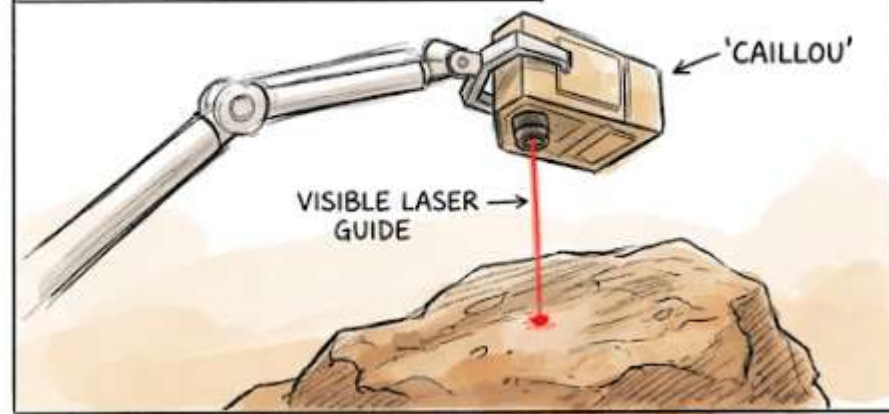


Mission Scenario

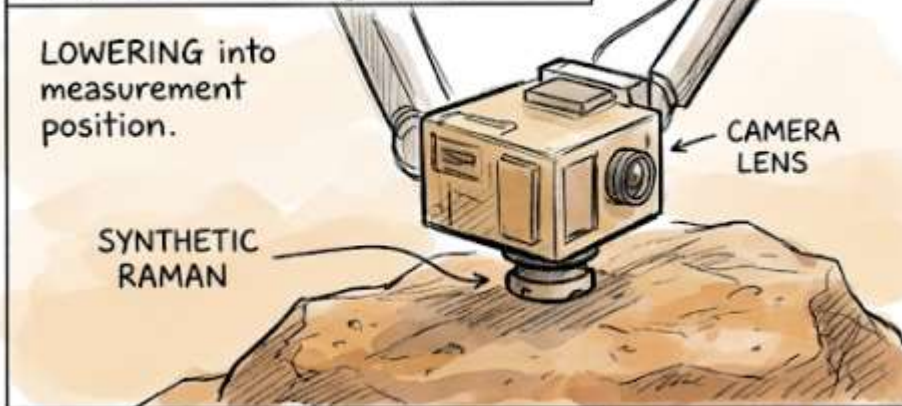
Etape 1: Picking up the device



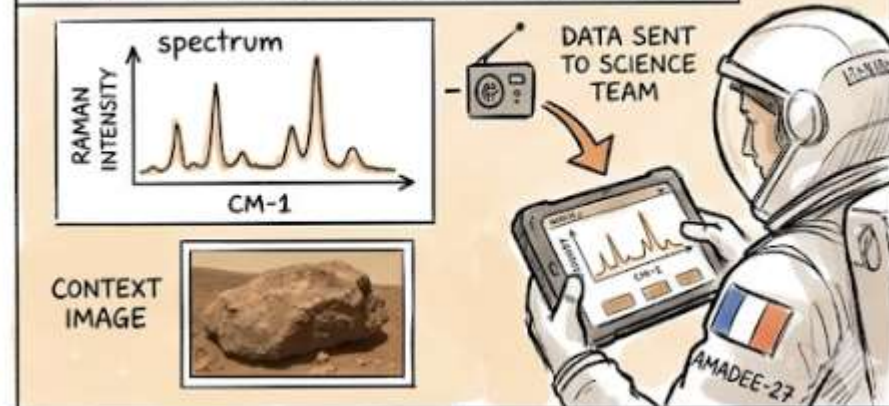
Etape 2: Target Acquisition



Etape 3: Lowering the sensor



Etape 4: Simulation & Transmission



Hardware & Mechanical Design

